

3. Convex functions

Outline

Convex functions

Operations that preserve convexity

Constructive convex analysis

Perspective and conjugate

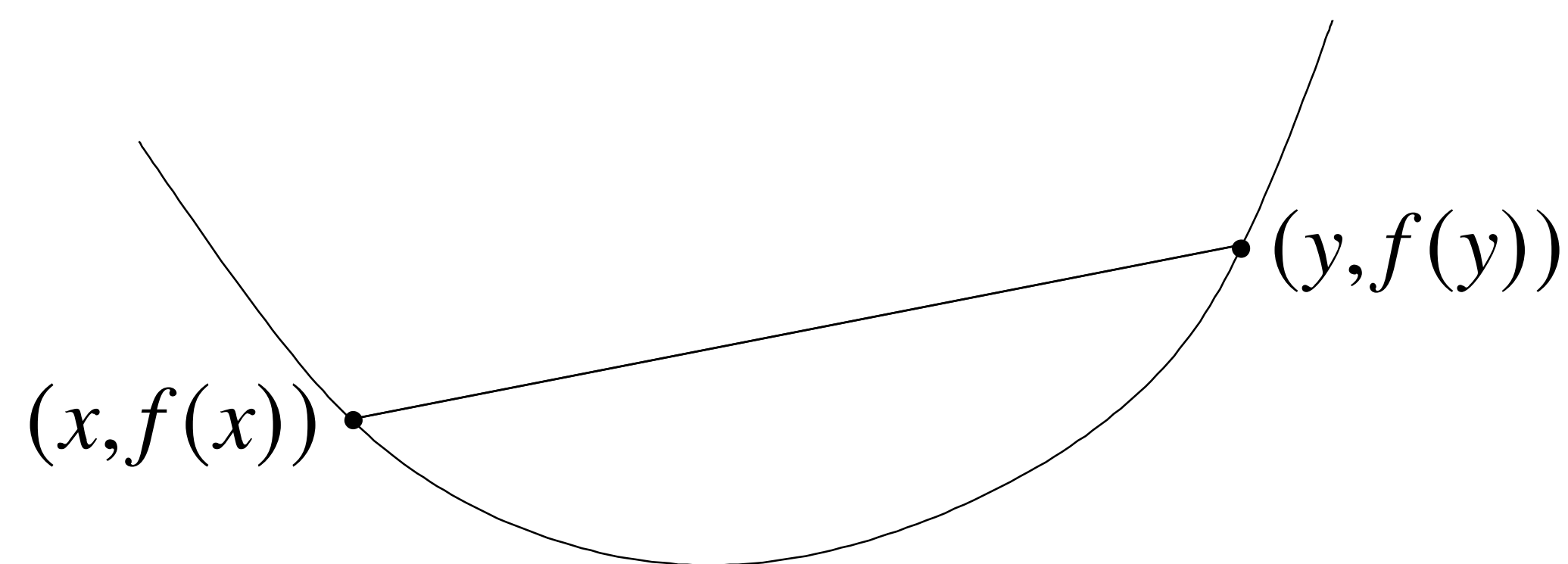
Quasiconvexity

Definition

Important!

- $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is convex if $\mathbf{dom} f$ is a convex set and for all $x, y \in \mathbf{dom} f$, $0 \leq \theta \leq 1$,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$



- f is concave if $-f$ is convex
- f is strictly convex if $\mathbf{dom} f$ is convex and for $x, y \in \mathbf{dom} f$, $x \neq y$, $0 < \theta < 1$,

$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$$

Examples on $\mathbf{R}$

convex functions:

- ▶ affine: $ax + b$ on $\mathbf{R}$, for any $a, b \in \mathbf{R}$
- ▶ exponential: e^{ax} , for any $a \in \mathbf{R}$
- ▶ powers: x^α on $\mathbf{R}_{++}$, for $\alpha \geq 1$ or $\alpha \leq 0$
- ▶ powers of absolute value: $|x|^p$ on $\mathbf{R}$, for $p \geq 1$
- ▶ positive part (relu): $\max\{0, x\}$

concave functions:

- ▶ affine: $ax + b$ on $\mathbf{R}$, for any $a, b \in \mathbf{R}$
- ▶ powers: x^α on $\mathbf{R}_{++}$, for $0 \leq \alpha \leq 1$
- ▶ logarithm: $\log x$ on $\mathbf{R}_{++}$
- ▶ entropy: $-x \log x$ on $\mathbf{R}_{++}$
- ▶ negative part: $\min\{0, x\}$

Examples on $\mathbf{R}^n$

convex functions:

- ▶ affine functions: $f(x) = a^T x + b$
- ▶ any norm, e.g., the ℓ_p norms
 - $\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p}$ for $p \geq 1$
 - $\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$
- ▶ sum of squares: $\|x\|_2^2 = x_1^2 + \dots + x_n^2$
- ▶ max function: $\max(x) = \max\{x_1, x_2, \dots, x_n\}$
- ▶ softmax or log-sum-exp function: $\log(\exp x_1 + \dots + \exp x_n)$

Triangle ineq: $\|x+y\| \leq \|x\| + \|y\|$.

Scalar mult: $\|\alpha x\| = \alpha \|x\|$.

$$\begin{aligned} \max(x) &\leq \text{LSE}(x) \leq \max(x) + \log(n) \\ \left(e^{\max(x)} &\leq \sum_i e^{x_i} \leq n e^{\max(x)} \right) \quad \frac{\partial \text{LSE}}{\partial x_i} = \frac{e^{x_i}}{e^{x_1} + \dots + e^{x_n}} \end{aligned}$$

Examples on $\mathbf{R}^{m \times n}$

- ▶ $X \in \mathbf{R}^{m \times n}$ ($m \times n$ matrices) is the variable
- ▶ general affine function has form

$$f(X) = \underbrace{\mathbf{tr}(A^T X)}_{\text{treating } A, X \text{ like vectors.}} + b = \sum_{i=1}^m \sum_{j=1}^n A_{ij} X_{ij} + b$$

for some $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}$

- ▶ spectral norm (maximum singular value) is convex

$$f(X) = \|X\|_2 = \sigma_{\max}(X) = (\lambda_{\max}(X^T X))^{1/2}$$

$$\sigma_{\max}(X) = \max_{\substack{\|x\|_2 \leq 1 \\ \|y\|_2 \leq 1}} x^T X y$$

- ▶ log-determinant: for $X \in \mathbf{S}_{++}^n$, $f(X) = \log \det X$ is concave

Extended-value extension

- ▶ suppose f is convex on $\mathbf{R}^n$, with domain $\mathbf{dom} f$
- ▶ its extended-value extension $\tilde{f}$ is function $\tilde{f} : \mathbf{R}^n \rightarrow \mathbf{R} \cup \{\infty\}$

$$\tilde{f}(x) = \begin{cases} f(x) & x \in \mathbf{dom} f \\ \infty & x \notin \mathbf{dom} f \end{cases}$$

- ▶ often simplifies notation; for example, the condition

$$0 \leq \theta \leq 1 \quad \implies \quad \tilde{f}(\theta x + (1 - \theta)y) \leq \theta \tilde{f}(x) + (1 - \theta) \tilde{f}(y)$$

(as an inequality in $\mathbf{R} \cup \{\infty\}$), means the same as the two conditions

- $\mathbf{dom} f$ is convex
- $x, y \in \mathbf{dom} f, 0 \leq \theta \leq 1 \implies f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$

Restriction of a convex function to a line

- ▶ $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is convex if and only if the function $g : \mathbf{R} \rightarrow \mathbf{R}$,

$$g(t) = f(x + tv), \quad \mathbf{dom} \, g = \{t \mid x + tv \in \mathbf{dom} f\}$$

is convex (in t) for any $x \in \mathbf{dom} f$, $v \in \mathbf{R}^n$

- ▶ can check convexity of f by checking convexity of functions of one variable

Example

- ▶ $f : \mathbf{S}^n \rightarrow \mathbf{R}$ with $f(X) = \log \det X$, $\text{dom } f = \mathbf{S}_{++}^n$ *direction.*
- ▶ consider line in $\mathbf{S}^n$ given by $X + tV$, $X \in \mathbf{S}_{++}^n$, $V \in \mathbf{S}^n$, $t \in \mathbf{R}$ } s.t. $X + tV \in \mathbf{S}_{++}^n$.

$$\begin{aligned}
 g(t) &= \log \det(X + tV) \\
 &= \log \det \left(X^{1/2} \left(I + tX^{-1/2} V X^{-1/2} \right) X^{1/2} \right) \\
 &= \log \det X + \log \det \left(I + tX^{-1/2} V X^{-1/2} \right) \quad \text{Recall def of char. poly.} \\
 &= \log \det X + \sum_{i=1}^n \log(1 + t\lambda_i) \quad \begin{aligned} \mathcal{P}_A(s) &= \det(sI - A) = \prod_{i=1}^n (s - \lambda_i) \\ \Leftrightarrow \det(I + A) &= \prod_{i=1}^n (1 + \lambda_i) \end{aligned}
 \end{aligned}$$

where λ_i are the eigenvalues of $X^{-1/2} V X^{-1/2}$

- ▶ g is concave in t (for any choice of $X \in \mathbf{S}_{++}^n$, $V \in \mathbf{S}^n$); hence f is concave

First-order condition

- ▶ f is **differentiable** if $\text{dom } f$ is open and the gradient

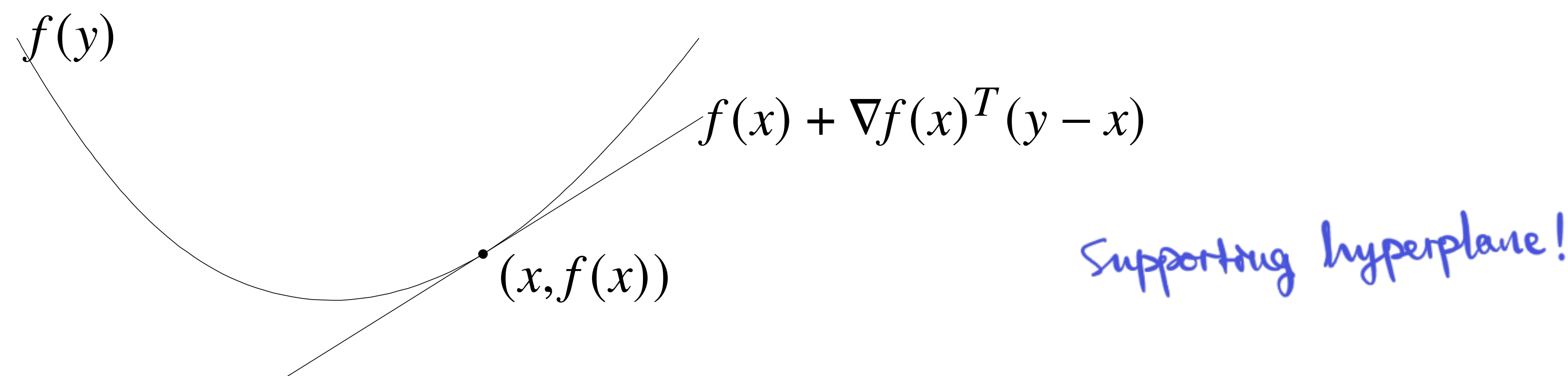
$$\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right) \in \mathbf{R}^n$$

exists at each $x \in \text{dom } f$

- ▶ **1st-order condition:** differentiable f with convex domain is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in \text{dom } f$$

- ▶ first order Taylor approximation of convex f is a **global underestimator** of f



Second-order conditions

- ▶ f is **twice differentiable** if $\mathbf{dom} f$ is open and the Hessian $\nabla^2 f(x) \in \mathbf{S}^n$,

$$\nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}, \quad i, j = 1, \dots, n,$$

exists at each $x \in \mathbf{dom} f$

- ▶ **2nd-order conditions:** for twice differentiable f with convex domain
 - f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in \mathbf{dom} f$
 - if $\nabla^2 f(x) \succ 0$ for all $x \in \mathbf{dom} f$, then f is strictly convex

Examples

- **quadratic function:** $f(x) = (1/2)x^T Px + q^T x + r$ (with $P \in \mathbf{S}^n$)

$$\nabla f(x) = Px + q, \quad \nabla^2 f(x) = P$$

convex if $P \geq 0$ (concave if $P \leq 0$)

- **least-squares objective:** $f(x) = \|Ax - b\|_2^2$

$$\nabla f(x) = 2A^T(Ax - b), \quad \nabla^2 f(x) = 2A^T A$$

convex (for any A)

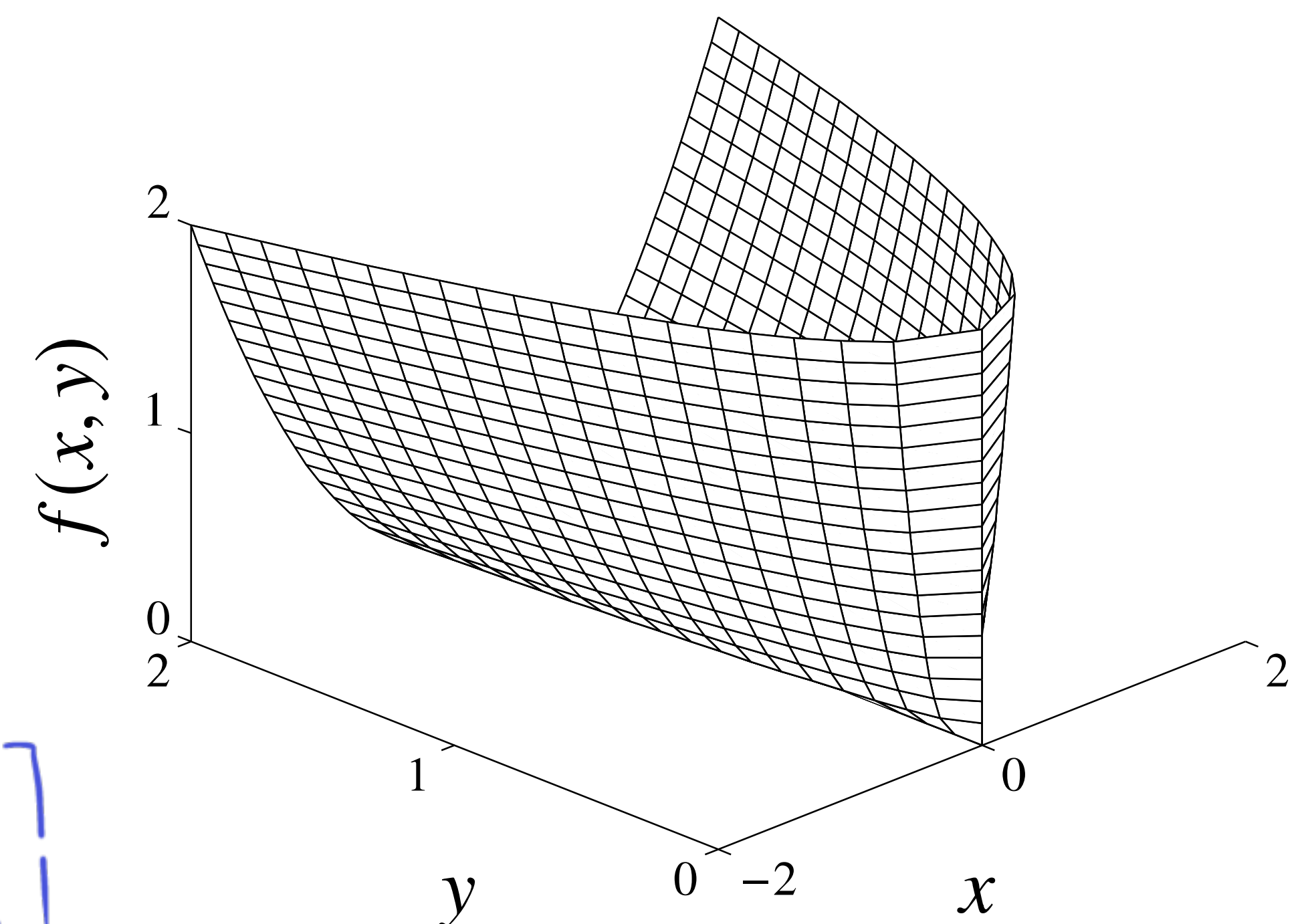
- **quadratic-over-linear:** $f(x, y) = x^2/y, y > 0$

$$\nabla^2 f(x, y) = \frac{2}{y^3} \begin{bmatrix} y \\ -x \end{bmatrix} \begin{bmatrix} y \\ -x \end{bmatrix}^T \geq 0$$

convex for $y > 0$

$$\nabla f = \begin{bmatrix} 2x/y \\ -x^2/y^2 \end{bmatrix}, \quad \nabla^2 f = \begin{bmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & -2\frac{x^2}{y^3} \end{bmatrix}$$

Boyd and Vandenberghe



More examples

$$\frac{\partial f}{\partial x_k} = \frac{\partial f}{\partial z_k} \frac{\partial z_k}{\partial x_k} = z_k \frac{\partial f}{\partial z_k} = \frac{z_k}{\sum_k z_k}$$

$$\frac{\partial^2 f}{\partial x_{k'} \partial x_k} = \frac{\partial}{\partial x_{k'}} \left(\frac{\partial f}{\partial x_k} \right) = z_{k'} \frac{\partial}{\partial z_{k'}} \left(\frac{z_k}{\sum z_k} \right) = z_{k'} \left(\delta_{kk'} \frac{1}{\sum z_k} - \frac{z_k}{(\sum z_k)^2} \right).$$

- **log-sum-exp:** $f(x) = \log \sum_{k=1}^n \exp x_k$ is convex

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T z} \mathbf{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T \quad (z_k = \exp x_k)$$

- to show $\nabla^2 f(x) \succeq 0$, we must verify that $v^T \nabla^2 f(x) v \geq 0$ for all v :

$$v^T \nabla^2 f(x) v = \frac{(\sum_k z_k v_k^2)(\sum_k z_k) - (\sum_k v_k z_k)^2}{(\sum_k z_k)^2} \geq 0$$

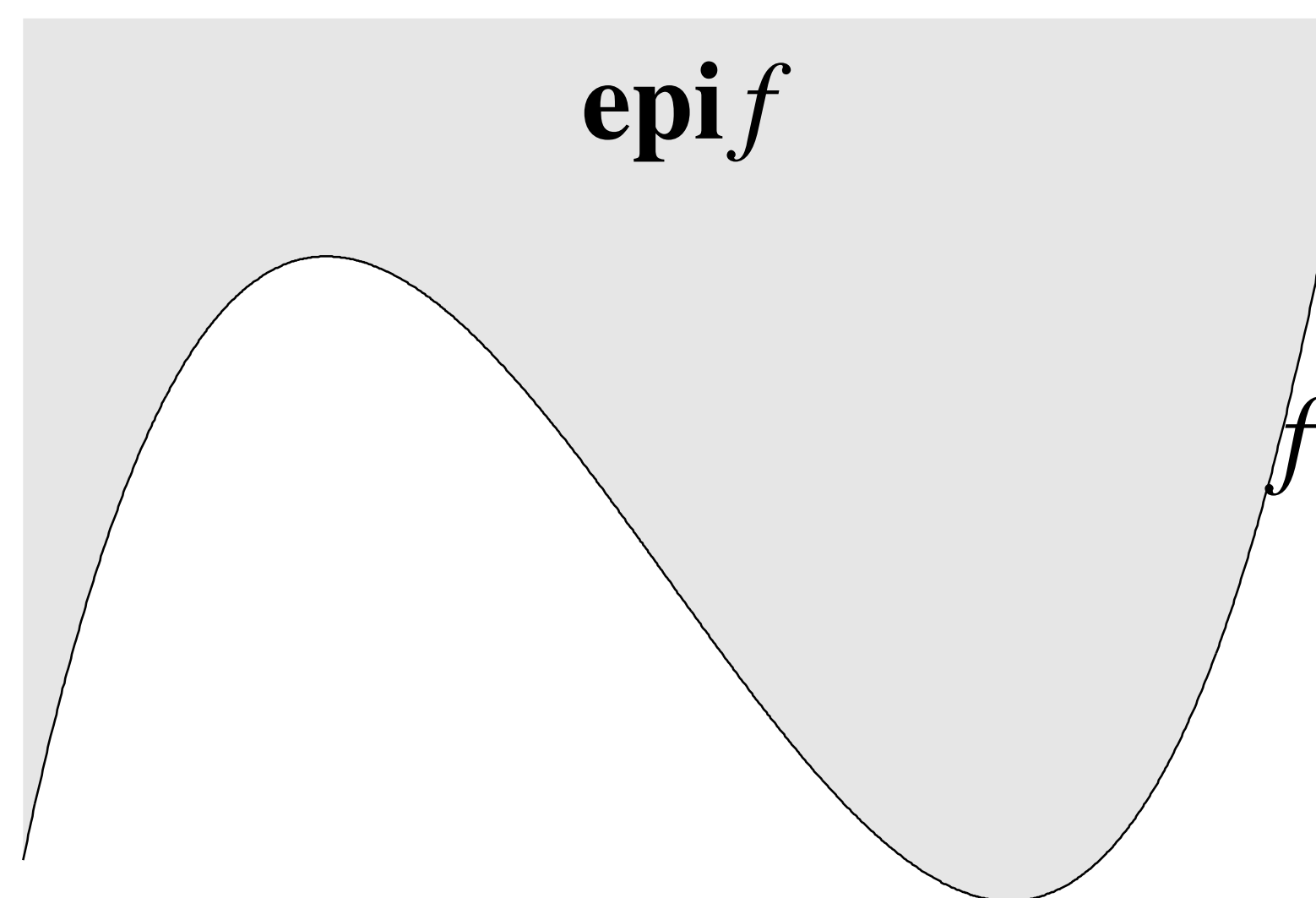
since $(\sum_k v_k z_k)^2 \leq (\sum_k z_k v_k^2)(\sum_k z_k)$ (from Cauchy-Schwarz inequality)

$$z_k > 0, \quad \lambda_k = \frac{z_k}{\sum z_k}, \quad \left(\sum_k \lambda_k v_k^2 \right) \leq \sum_k \lambda_k v_k^2$$

- **geometric mean:** $f(x) = (\prod_{k=1}^n x_k)^{1/n}$ on $\mathbf{R}_{++}^n$ is concave (similar proof as above)

Epigraph and sublevel set

- ▶ α -**sublevel set** of $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is $C_\alpha = \{x \in \mathbf{dom} f \mid f(x) \leq \alpha\}$
- ▶ sublevel sets of convex functions are convex sets (but converse is false)
- ▶ **epigraph** of $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is $\mathbf{epi} f = \{(x, t) \in \mathbf{R}^{n+1} \mid x \in \mathbf{dom} f, f(x) \leq t\}$



- ▶ f is convex if and only if $\mathbf{epi} f$ is a convex set

Jensen's inequality

- ▶ **basic inequality:** if f is convex, then for $x, y \in \mathbf{dom} f$, $0 \leq \theta \leq 1$,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

- ▶ **extension:** if f is convex and z is a random variable on $\mathbf{dom} f$,

$$f(\mathbf{E} z) \leq \mathbf{E} f(z)$$

- ▶ basic inequality is special case with discrete distribution

$$\mathbf{prob}(z = x) = \theta, \quad \mathbf{prob}(z = y) = 1 - \theta$$

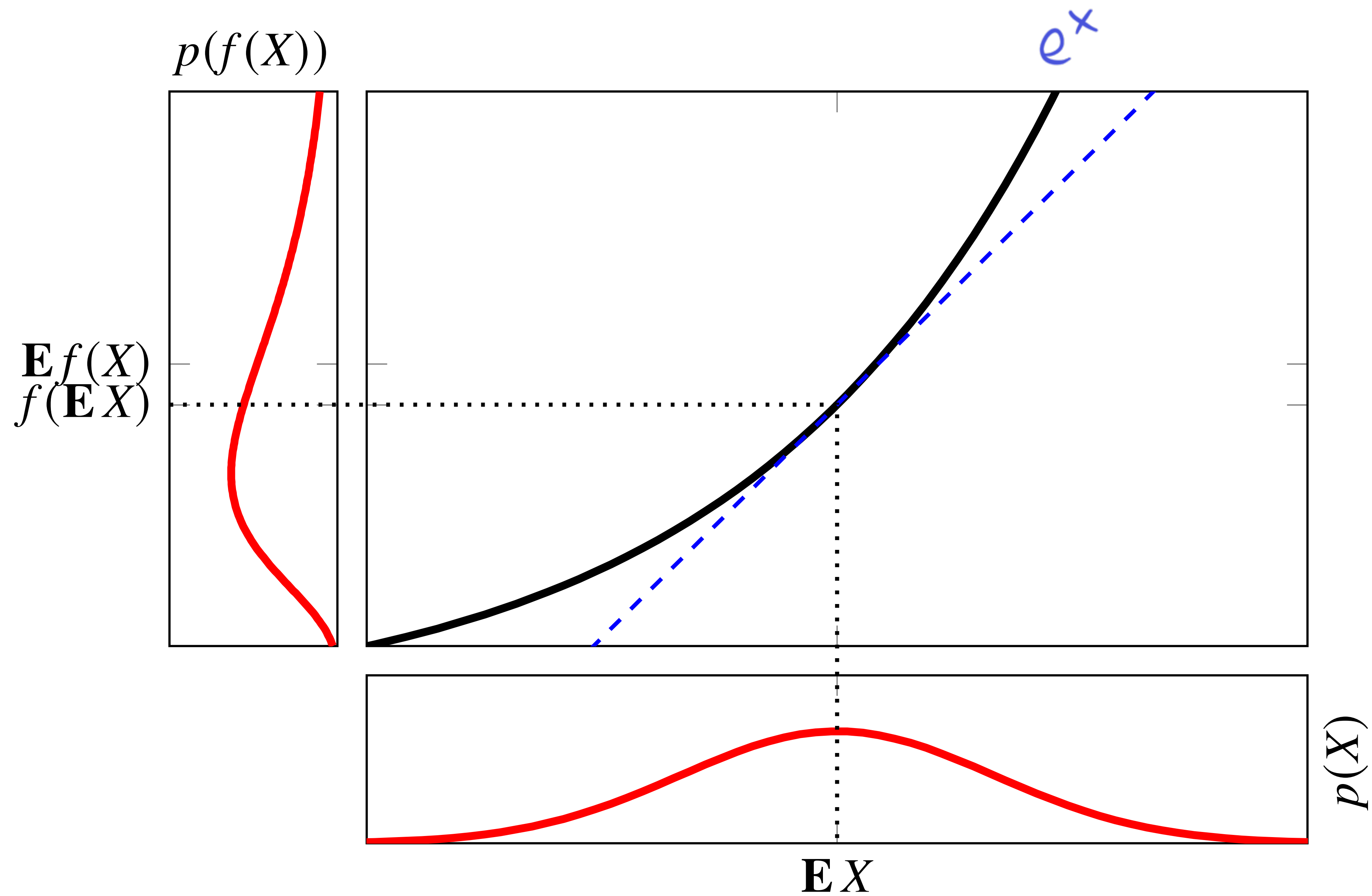
Example: log-normal random variable

- ▶ suppose $X \sim \mathcal{N}(\mu, \sigma^2)$
- ▶ with $f(u) = \exp u$, $Y = f(X)$ is log-normal
- ▶ we have $\mathbf{E}f(X) = \exp(\mu + \sigma^2/2)$
- ▶ Jensen's inequality is

$$f(\mathbf{E} X) = \exp \mu \leq \mathbf{E}f(X) = \exp(\mu + \sigma^2/2)$$

which indeed holds since $\exp \sigma^2/2 > 1$

Example: log-normal random variable



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Showing a function is convex

methods for establishing convexity of a function f

1. verify definition (often simplified by restricting to a line)
2. for twice differentiable functions, show $\nabla^2 f(x) \succeq 0$
 - recommended only for **very simple** functions
3. show that f is obtained from simple convex functions by operations that preserve convexity
 - nonnegative weighted sum
 - composition with affine function
 - pointwise maximum and supremum
 - composition
 - minimization
 - perspective

you'll mostly use methods 2 and 3

Nonnegative scaling, sum, and integral

- ▶ **nonnegative multiple:** αf is convex if f is convex, $\alpha \geq 0$
- ▶ **sum:** $f_1 + f_2$ convex if f_1, f_2 convex
- ▶ **infinite sum:** if $f_1, f_2, \dots$ are convex functions, infinite sum $\sum_{i=1}^{\infty} f_i$ is convex
- ▶ **integral:** if $f(x, \alpha)$ is convex in x for each $\alpha \in \mathcal{A}$, then $\int_{\alpha \in \mathcal{A}} f(x, \alpha) d\alpha$ is convex
- ▶ there are analogous rules for concave functions

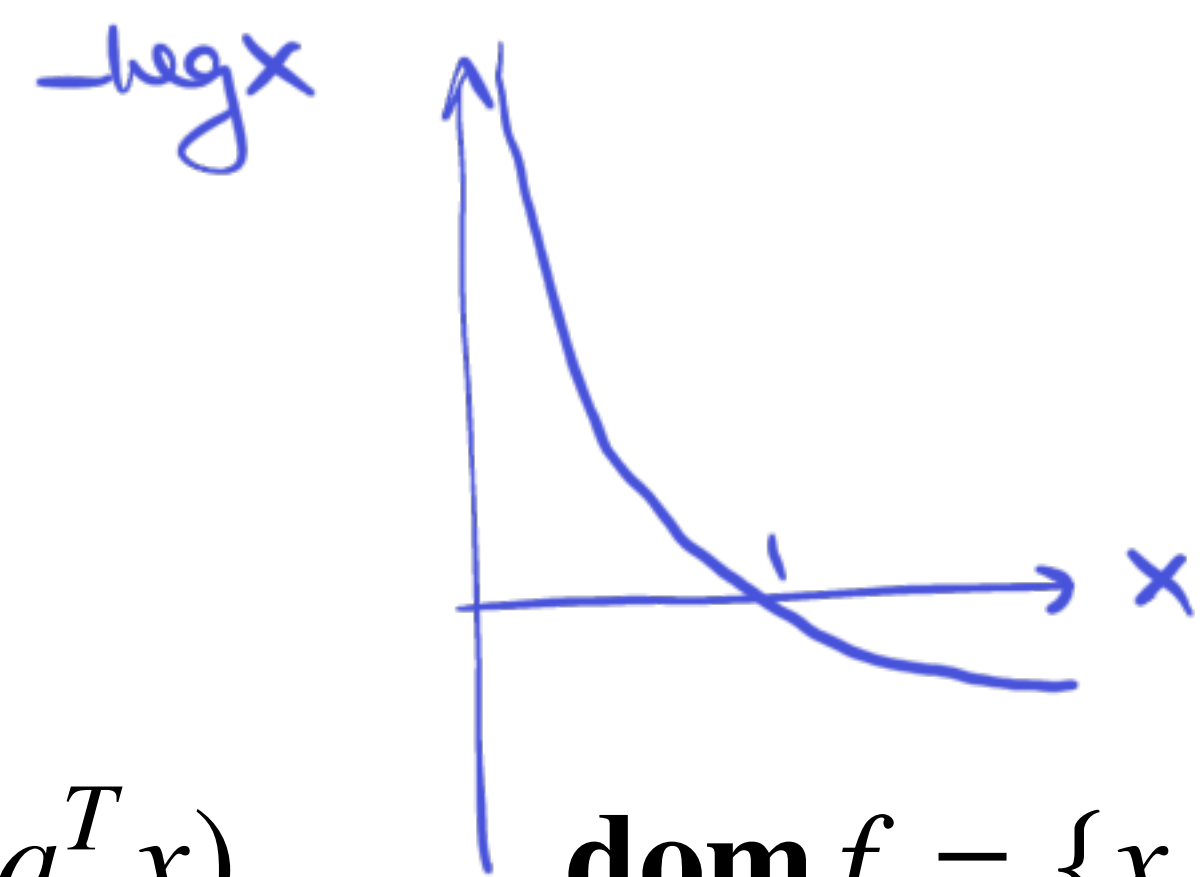
Composition with affine function

(pre-)composition with affine function: $f(Ax + b)$ is convex if f is convex

examples

- ▶ log barrier for linear inequalities

$$f(x) = - \sum_{i=1}^m \log(b_i - a_i^T x),$$

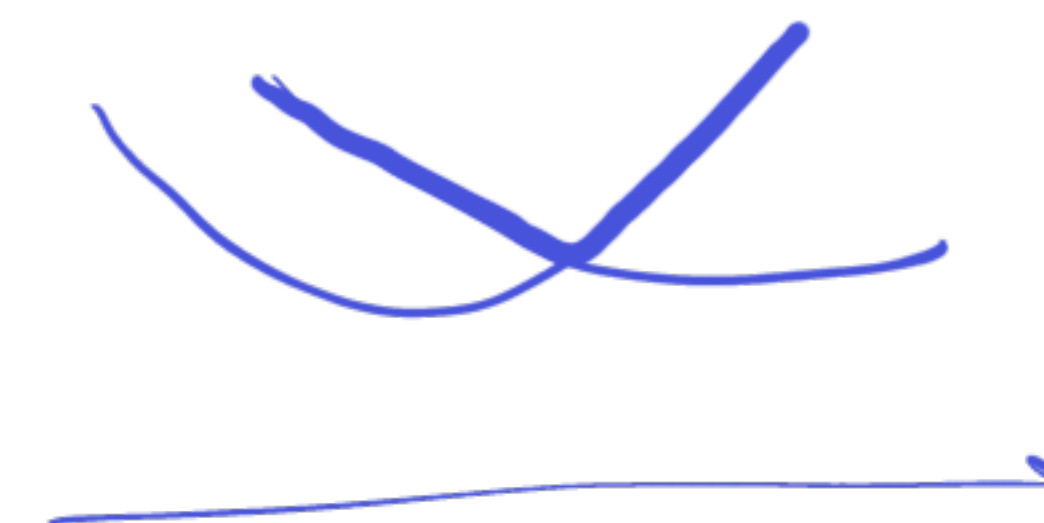


$$\text{dom } f = \{x \mid a_i^T x < b_i, i = 1, \dots, m\}$$

- ▶ norm approximation error: $f(x) = \|Ax - b\|$ (any norm)

Pointwise maximum

if $f_1, \dots, f_m$ are convex, then $f(x) = \max\{f_1(x), \dots, f_m(x)\}$ is convex



examples

- ▶ piecewise-linear function: $f(x) = \max_{i=1, \dots, m} (a_i^T x + b_i)$
- ▶ sum of r largest components of $x \in \mathbf{R}^n$:

$$f(x) = x_{[1]} + x_{[2]} + \dots + x_{[r]}$$

($x_{[i]}$ is i th largest component of x)

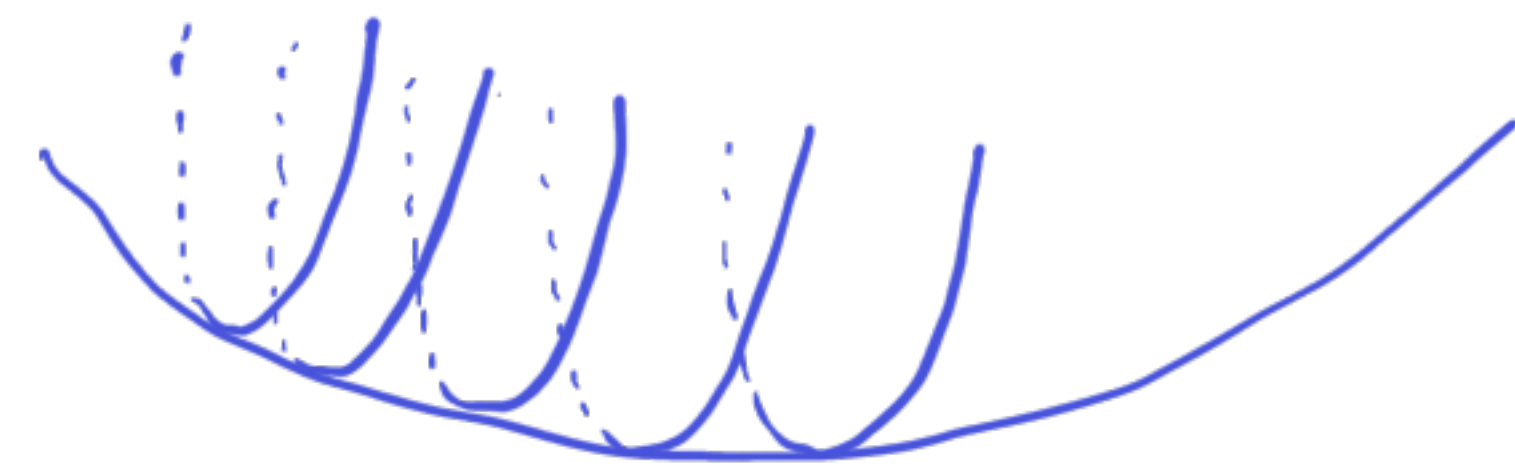
proof: $f(x) = \max\{x_{i_1} + x_{i_2} + \dots + x_{i_r} \mid 1 \leq i_1 < i_2 < \dots < i_r \leq n\}$

Pointwise supremum

if $f(x, y)$ is convex in x for each $y \in \mathcal{A}$, then $g(x) = \sup_{y \in \mathcal{A}} f(x, y)$ is convex

examples

- ▶ distance to farthest point in a set C : $f(x) = \sup_{y \in C} \|x - y\|$
- ▶ maximum eigenvalue of symmetric matrix: for $X \in \mathbf{S}^n$, $\lambda_{\max}(X) = \sup_{\|y\|_2=1} y^T X y$ is convex
- ▶ support function of a set C : $S_C(x) = \sup_{y \in C} y^T x$ is convex



Partial minimization

- ▶ the function $g(x) = \inf_{y \in C} f(x, y)$ is called the **partial minimization** of f (w.r.t. y)
- ▶ if $f(x, y)$ is convex in (x, y) and C is a convex set, then partial minimization g is convex

examples

- ▶ $f(x, y) = x^T A x + 2x^T B y + y^T C y$ with

$$\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \succeq 0, \quad C \succ 0$$

minimizing over y gives $g(x) = \inf_y f(x, y) = x^T (A - B C^{-1} B^T) x$
 g is convex, hence Schur complement $A - B C^{-1} B^T \succeq 0$

- ▶ distance to a set: **dist** $(x, S) = \inf_{y \in S} \|x - y\|$ is convex if S is convex

$$\begin{aligned} & a x^2 + 2 b x y + c y^2, \quad c > 0. \\ & y = -\frac{b x}{c} \text{ is minimization.} \\ & a x^2 - 2 \frac{b^2 x^2}{c} + \frac{b^2 x^2}{c} \\ & = (a - b^2/c) x^2. \end{aligned}$$

Composition with scalar functions

- ▶ composition of $g : \mathbf{R}^n \rightarrow \mathbf{R}$ and $h : \mathbf{R} \rightarrow \mathbf{R}$ is $f(x) = h(g(x))$ (written as $f = h \circ g$)
- ▶ composition f is convex if
 - g convex, h convex, $\tilde{h}$ nondecreasing
 - or g concave, h convex, $\tilde{h}$ nonincreasing(monotonicity must hold for extended-value extension $\tilde{h}$)
- ▶ proof (for $n = 1$, differentiable g, h)

$$f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x)$$

$$f'(x) = h'(g(x))g'(x)$$

examples

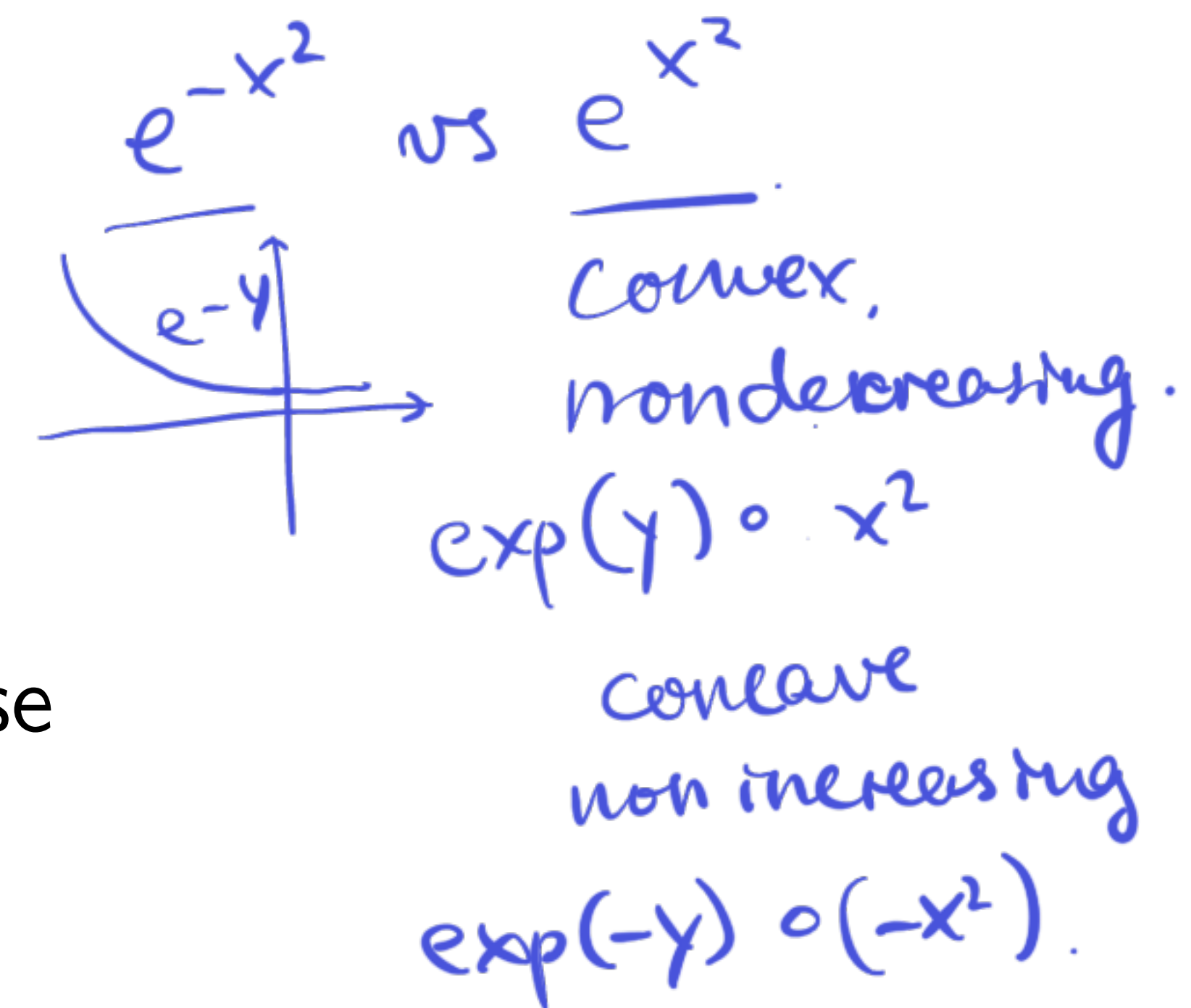
- ▶ $f(x) = \exp g(x)$ is convex if g is convex
- ▶ $f(x) = 1/g(x)$ is convex if g is concave and positive

e.g. $-\frac{1}{\|x\|}$

Nonconvex
example: e^{-x^2}

General composition rule

- ▶ composition of $g : \mathbf{R}^n \rightarrow \mathbf{R}^k$ and $h : \mathbf{R}^k \rightarrow \mathbf{R}$ is $f(x) = h(g(x)) = h(g_1(x), g_2(x), \dots, g_k(x))$
- ▶ f is convex if h is convex and for each i one of the following holds
 - g_i convex, $\tilde{h}$ nondecreasing in its i th argument
 - g_i concave, $\tilde{h}$ nonincreasing in its i th argument
 - g_i affine
- ▶ you will use this composition rule **constantly** throughout this course
- ▶ you need to commit this rule to memory



Examples

► $\log \sum_{i=1}^m \exp g_i(x)$ is convex if g_i are convex

(Recall. LogSumExp's derivatives are softmax-arg)
 $\frac{e^{x_i}}{\sum_j e^{x_j}} > 0$

► $f(x) = p(x)^2 / q(x)$ is convex if

- p is nonnegative and convex
- q is positive and concave

$\frac{x^2}{y}$ — nondecreasing if $x \geq 0, y > 0$

► composition rule subsumes others, e.g.,

- αf is convex if f is, and $\alpha \geq 0$
- sum of convex (concave) functions is convex (concave)
- max of convex functions is convex
- min of concave functions is concave

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Constructive convexity verification

- ▶ start with function f given as **expression**
- ▶ build parse tree for expression
 - leaves are variables or constants
 - nodes are functions of child expressions
- ▶ use composition rule to tag subexpressions as convex, concave, affine, or none
- ▶ if root node is labeled convex (concave), then f is convex (concave)
- ▶ extension: tag sign of each expression, and use sign-dependent monotonicity

- ▶ this is sufficient to show f is convex (concave), but not necessary
- ▶ this method for checking convexity (concavity) is readily automated

Example

the function

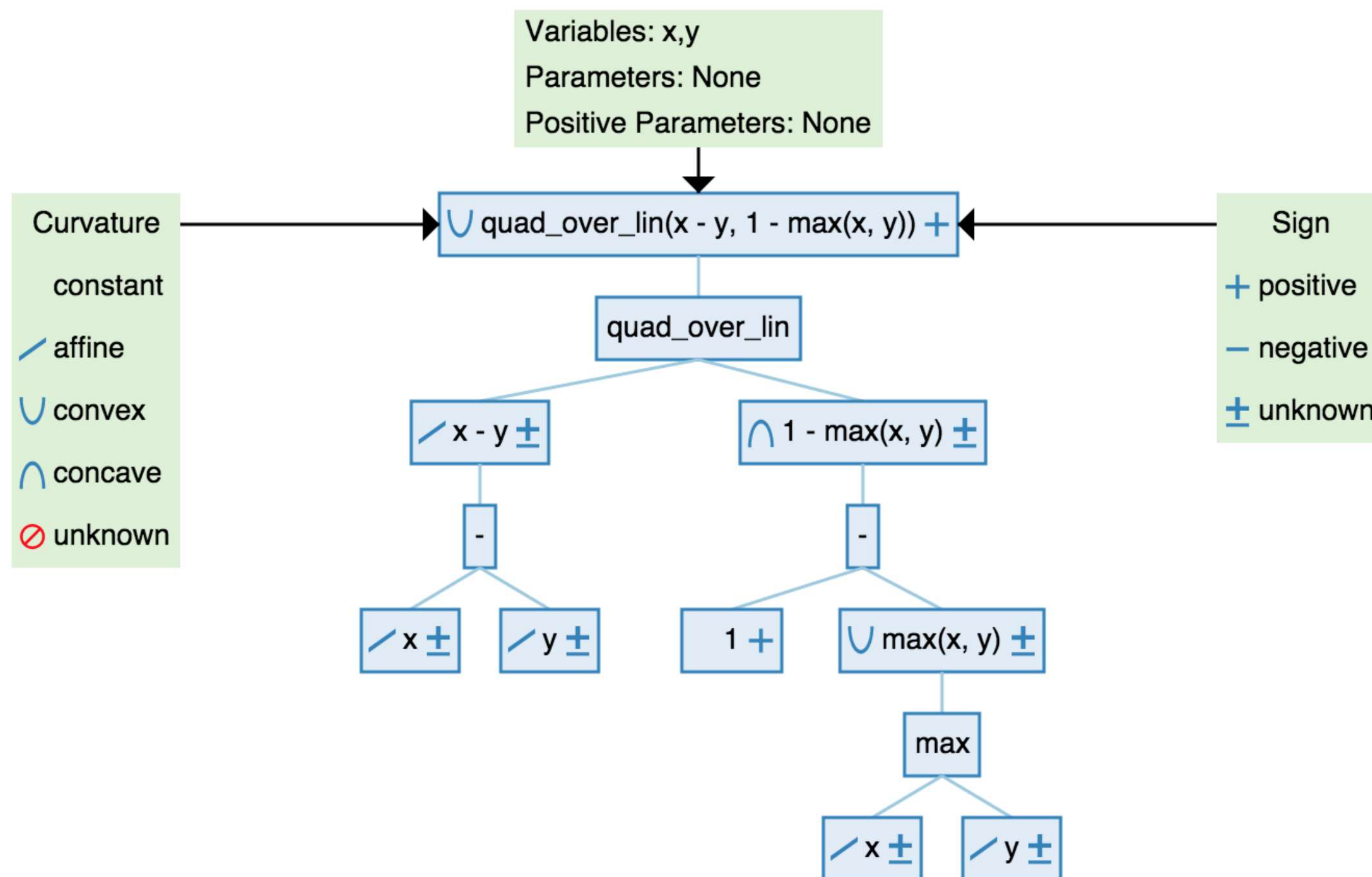
$$f(x, y) = \frac{(x - y)^2}{1 - \max(x, y)}, \quad x < 1, \quad y < 1$$

is convex

constructive analysis:

- ▶ (leaves) x , y , and 1 are affine
- ▶ $\max(x, y)$ is convex; $x - y$ is affine
- ▶ $1 - \max(x, y)$ is concave
- ▶ function u^2/v is convex, monotone decreasing in v for $v > 0$
- ▶ f is composition of u^2/v with $u = x - y$, $v = 1 - \max(x, y)$, hence convex

Example (from dcp.stanford.edu)



Disciplined convex programming

in **disciplined convex programming** (DCP) users construct convex and concave functions as expressions using constructive convex analysis

- ▶ expressions formed from
 - **variables**,
 - **constants**,
 - and **atomic functions** from a library
- ▶ atomic functions have known convexity, monotonicity, and sign properties
- ▶ all subexpressions match general composition rule
- ▶ a valid DCP function is
 - convex-by-construction
 - ‘syntactically’ convex (can be checked ‘locally’)
- ▶ convexity depends only on attributes of atomic functions, not their meanings
 - e.g., could swap $\sqrt{\cdot}$ and $\sqrt[4]{\cdot}$, or $\exp \cdot$ and $(\cdot)_+$, since their attributes match

CVXPY example

$$\frac{(x - y)^2}{1 - \max(x, y)}, \quad x < 1, \quad y < 1$$

```
import cvxpy as cp
x = cp.Variable()
y = cp.Variable()
expr = cp.quad_over_lin(x - y, 1 - cp.maximum(x, y))
expr.curvature # Convex
expr.sign # Positive
expr.is_dcp() # True
```

(atom quad_over_lin(u,v) includes domain constraint v>0)

DCP is only sufficient

- ▶ consider convex function $f(x) = \sqrt{1+x^2}$
- ▶ expression `f1 = cp.sqrt(1+cp.square(x))` is **not** DCP
- ▶ expression `f2 = cp.norm2([1,x])` **is** DCP
- ▶ CVXPY will not recognize `f1` as convex, even though it represents a convex function

*sqrt is concave.
So can't check by comp. rule.*

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Perspective

- ▶ the **perspective** of a function $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is the function $g : \mathbf{R}^n \times \mathbf{R} \rightarrow \mathbf{R}$,

$$g(x, t) = tf(x/t), \quad \text{dom } g = \{(x, t) \mid x/t \in \text{dom } f, t > 0\}$$

- ▶ g is convex if f is convex

$$(x, t, s) \in \text{epi } g \Leftrightarrow tf(x/t) \leq s$$

$$\Leftrightarrow f(x/t) \leq s/t$$

$$\Leftrightarrow (x/t, s/t) \in \text{epi } f$$

$$\text{epi } g \xrightarrow{\text{perspective function}} \text{epi } f.$$

examples

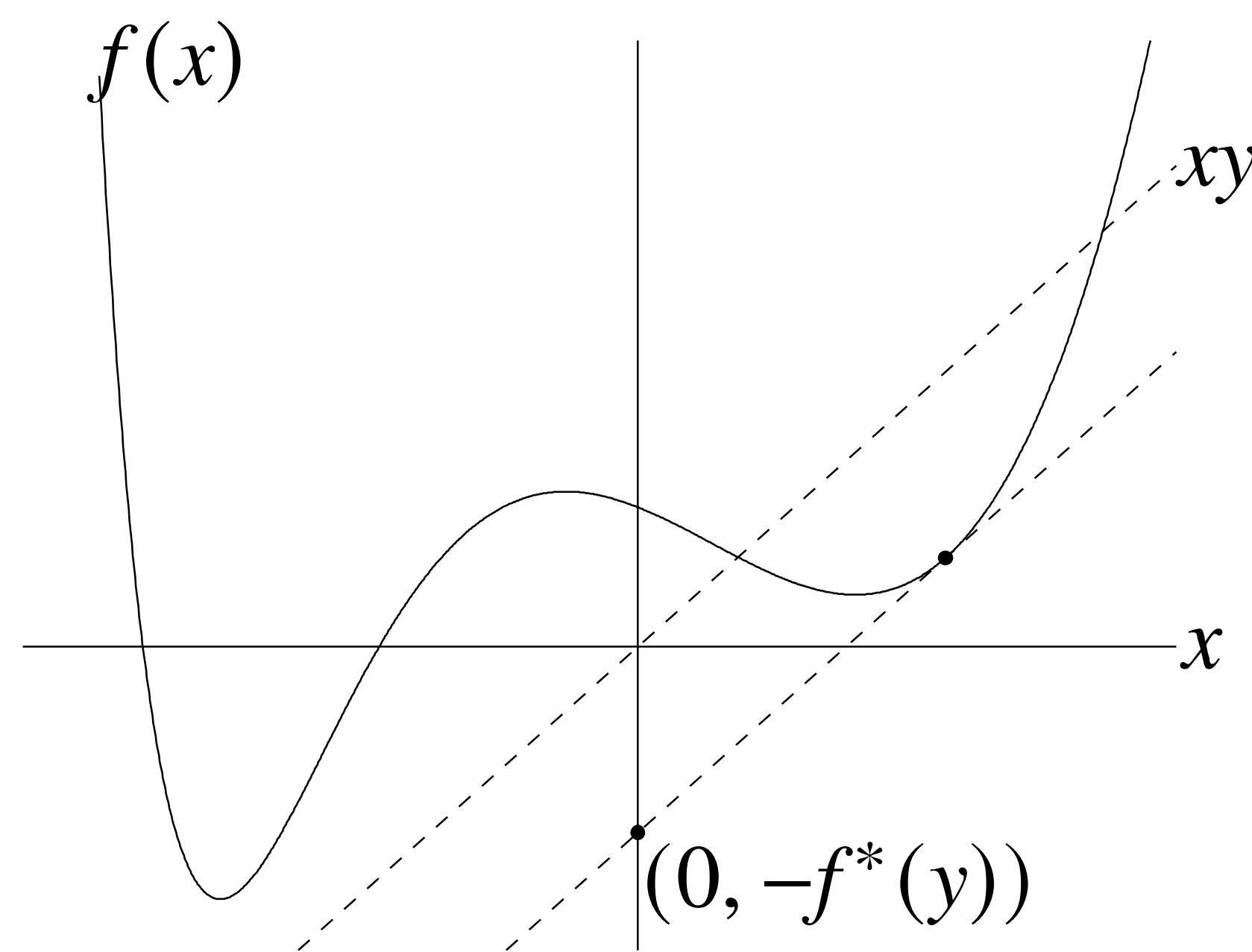
- ▶ $f(x) = x^T x$ is convex; so $g(x, t) = x^T x/t$ is convex for $t > 0$
- ▶ $f(x) = -\log x$ is convex; so relative entropy $g(x, t) = t \log t - t \log x$ is convex on $\mathbf{R}_{++}^2$

$$\begin{array}{c} \text{"} \\ -t \log \frac{x}{t} \end{array}$$

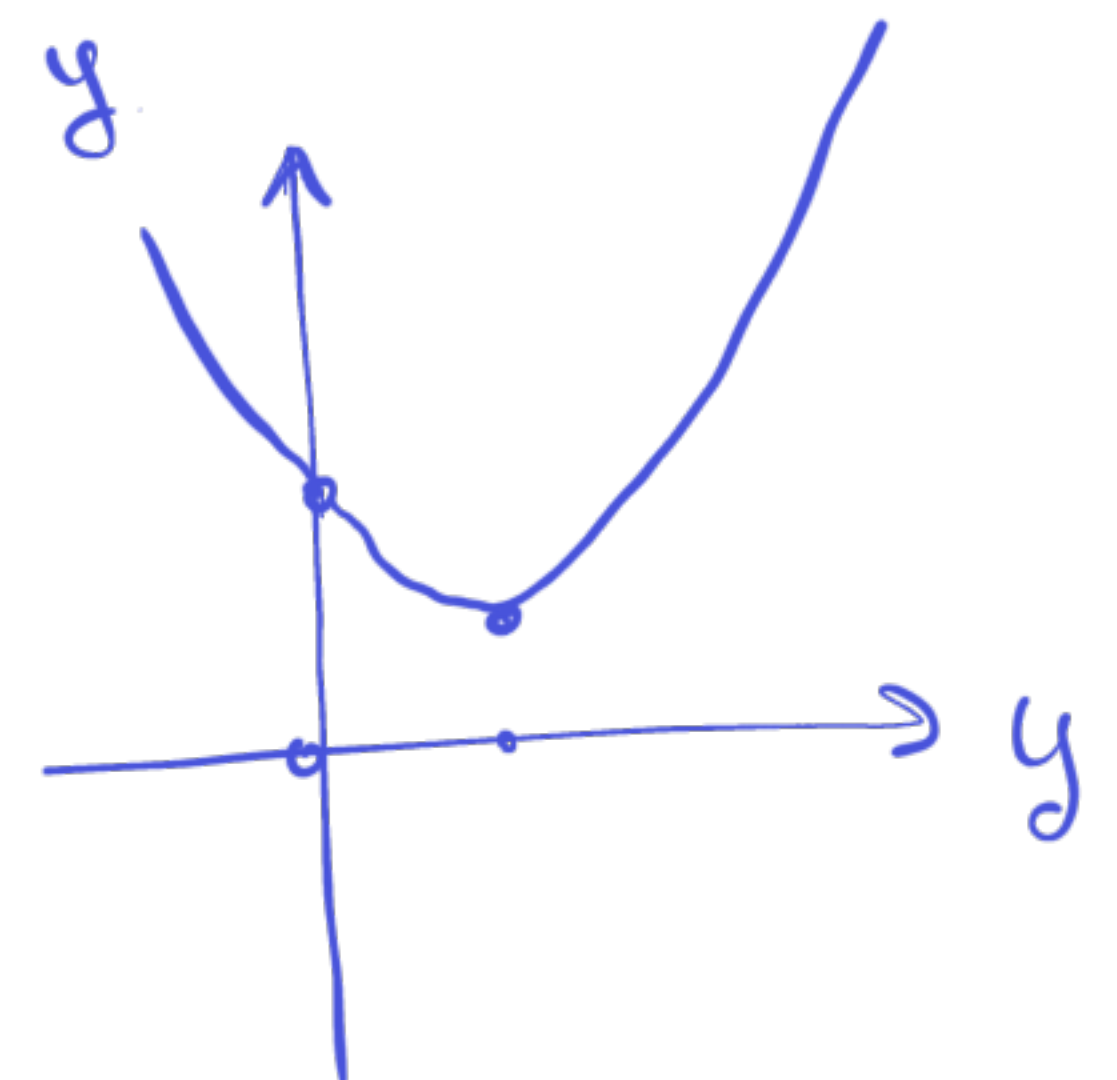
Conjugate function

$\equiv$ Legendre transform: thermodynamics...

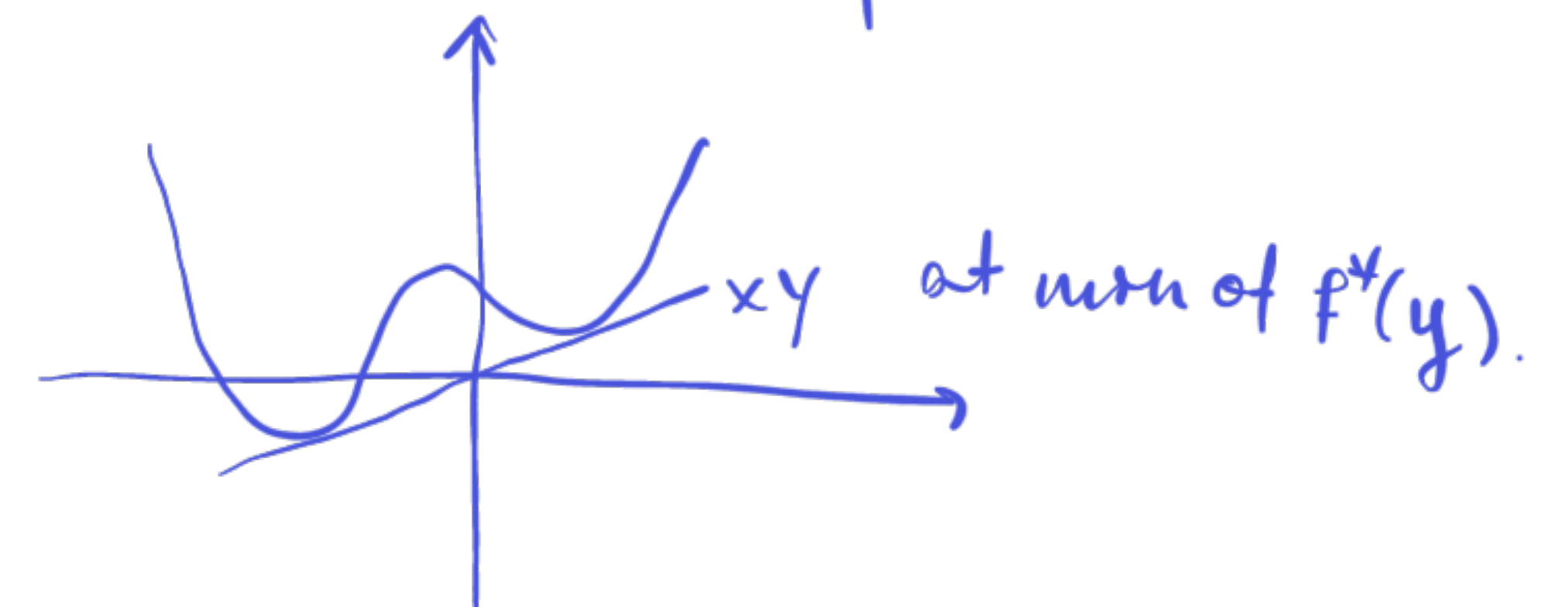
- ▶ the **conjugate** of a function f is $f^*(y) = \sup_{x \in \text{dom} f} (y^T x - f(x))$ } each is affine in y .



different const. y



- ▶ f^* is convex (even if f is not)
- ▶ will be useful in chapter 5



Examples

- ▶ negative logarithm $f(x) = -\log x$

$$f^*(y) = \sup_{x>0} (xy + \log x) = \begin{cases} -1 - \log(-y) & y < 0 \\ \infty & \text{otherwise} \end{cases}$$

$z = -xy > 0. \quad xy + \log x = \log \frac{z}{|y|} - z = \log z - z - \log |y|.$

$\frac{d}{dz} = \frac{1}{z} - 1.$

- ▶ strictly convex quadratic, $f(x) = (1/2)x^T Qx$ with $Q \in \mathbf{S}_{++}^n$

$$f^*(y) = \sup_x (y^T x - (1/2)x^T Qx) = \frac{1}{2}y^T Q^{-1}y$$

Outline

Convex functions

Operations that preserve convexity

Constructive convex analysis

Perspective and conjugate

Quasiconvexity

Quasiconvex functions

Generalization. Representable by a family of convex functions.

$$f(x) \leq t \Leftrightarrow \phi_t(x) \leq 0. \text{ convex.}$$

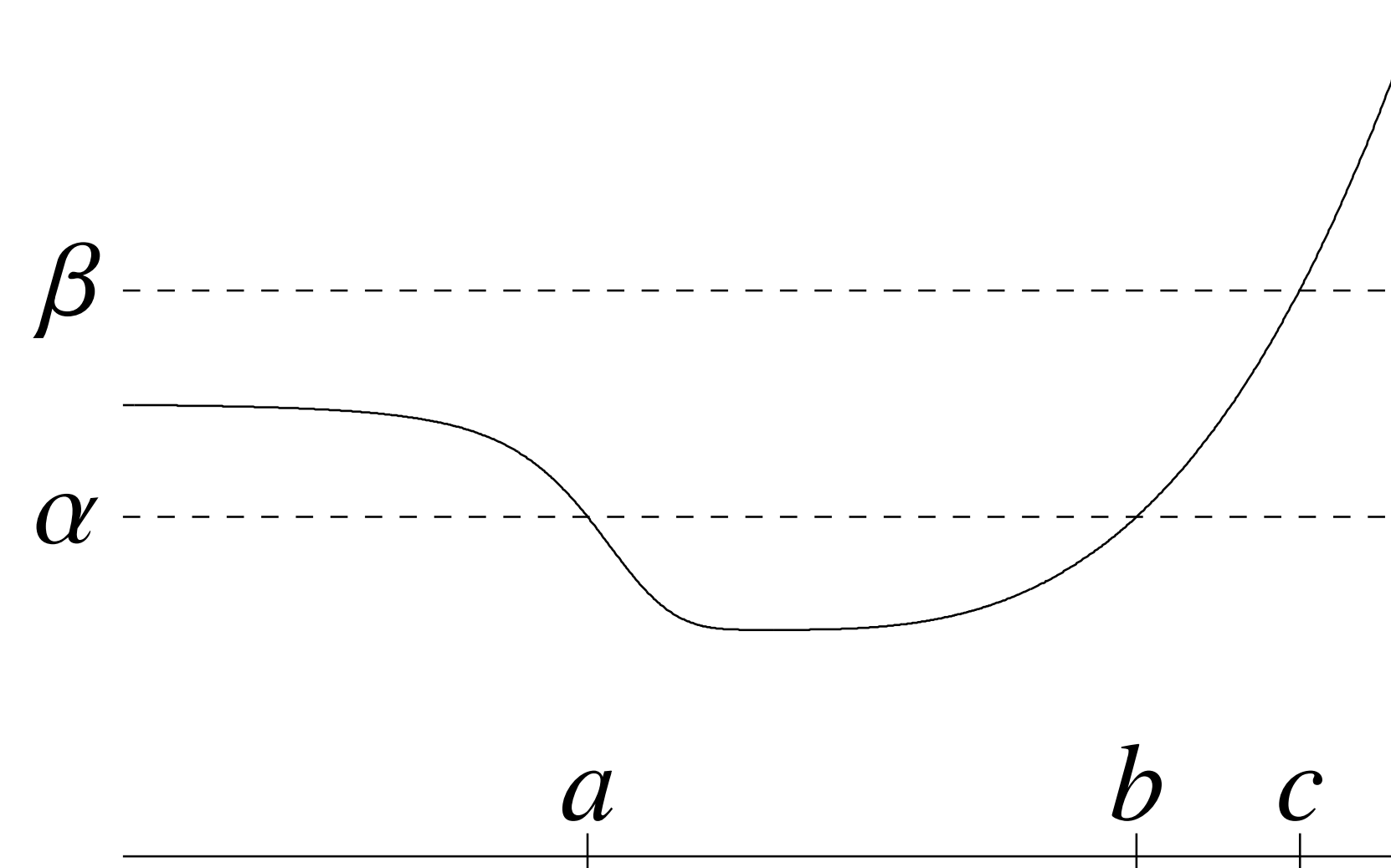
$$\text{e.g. } \phi_t(x) = \begin{cases} 0, & f(x) \leq t \\ \infty, & \text{else.} \end{cases}$$

$$\phi_t(x) = \text{dist}(x, \{z : f(z) \leq t\}).$$

- $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is **quasiconvex** if $\text{dom } f$ is convex and the sublevel sets

$$S_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$$

are convex for all α



- f is **quasiconcave** if $-f$ is quasiconvex
- f is **quasilinear** if it is quasiconvex and quasiconcave

Examples

- ▶ $\sqrt{|x|}$ is quasiconvex on $\mathbf{R}$
- ▶ $\text{ceil}(x) = \inf\{z \in \mathbf{Z} \mid z \geq x\}$ is quasilinear
- ▶ $\log x$ is quasilinear on $\mathbf{R}_{++}$
- ▶ $f(x_1, x_2) = x_1 x_2$ is quasiconcave on $\mathbf{R}_{++}^2$
- ▶ linear-fractional function

$$f(x) = \frac{a^T x + b}{c^T x + d}, \quad \text{dom } f = \{x \mid c^T x + d > 0\}$$

is quasilinear

Example: Internal rate of return

- ▶ cash flow $x = (x_0, \dots, x_n)$; x_i is payment in period i (to us if $x_i > 0$)
- ▶ we assume $x_0 < 0$ (*i.e.*, an initial investment) and $x_0 + x_1 + \dots + x_n > 0$
- ▶ **net present value** (NPV) of cash flow x , for interest rate r , is $PV(x, r) = \sum_{i=0}^n (1+r)^{-i} x_i$
- ▶ **internal rate of return** (IRR) is smallest interest rate for which $PV(x, r) = 0$:

$$IRR(x) = \inf\{r \geq 0 \mid PV(x, r) = 0\}$$

- ▶ IRR is quasiconcave: superlevel set is intersection of open halfspaces

$$IRR(x) \geq R \quad \Longleftrightarrow \quad \sum_{i=0}^n (1+r)^{-i} x_i > 0 \text{ for } 0 \leq r < R$$

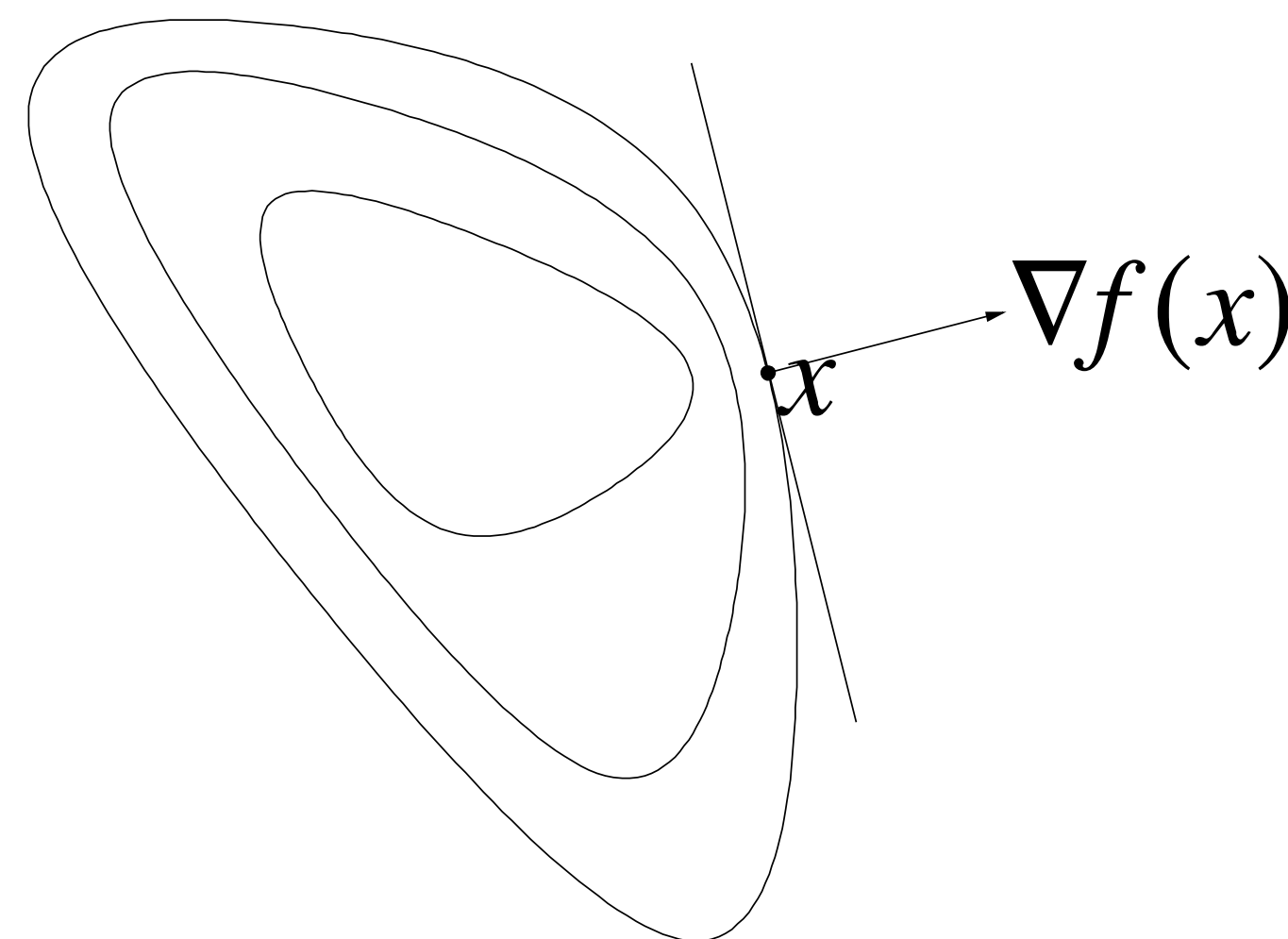
Properties of quasiconvex functions

- ▶ **modified Jensen inequality:** for quasiconvex f

$$0 \leq \theta \leq 1 \implies f(\theta x + (1 - \theta)y) \leq \max\{f(x), f(y)\}$$

- ▶ **first-order condition:** differentiable f with convex domain is quasiconvex if and only if

$$f(y) \leq f(x) \implies \nabla f(x)^T (y - x) \leq 0$$



- ▶ **sum** of quasiconvex functions is not necessarily quasiconvex